

03-Aug-2026

Introduction to Project Management

CE716 : Project Management and Controls

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Department of Civil Engineering



Agenda

- Difference between Project & Product
- What is a Project?
- Salient features of Construction project
- What is Construction Project Management?
- Life-cycle of a Construction project
- Project stakeholders
- Planning vs Scheduling



Construction Project Management

What?

Why?

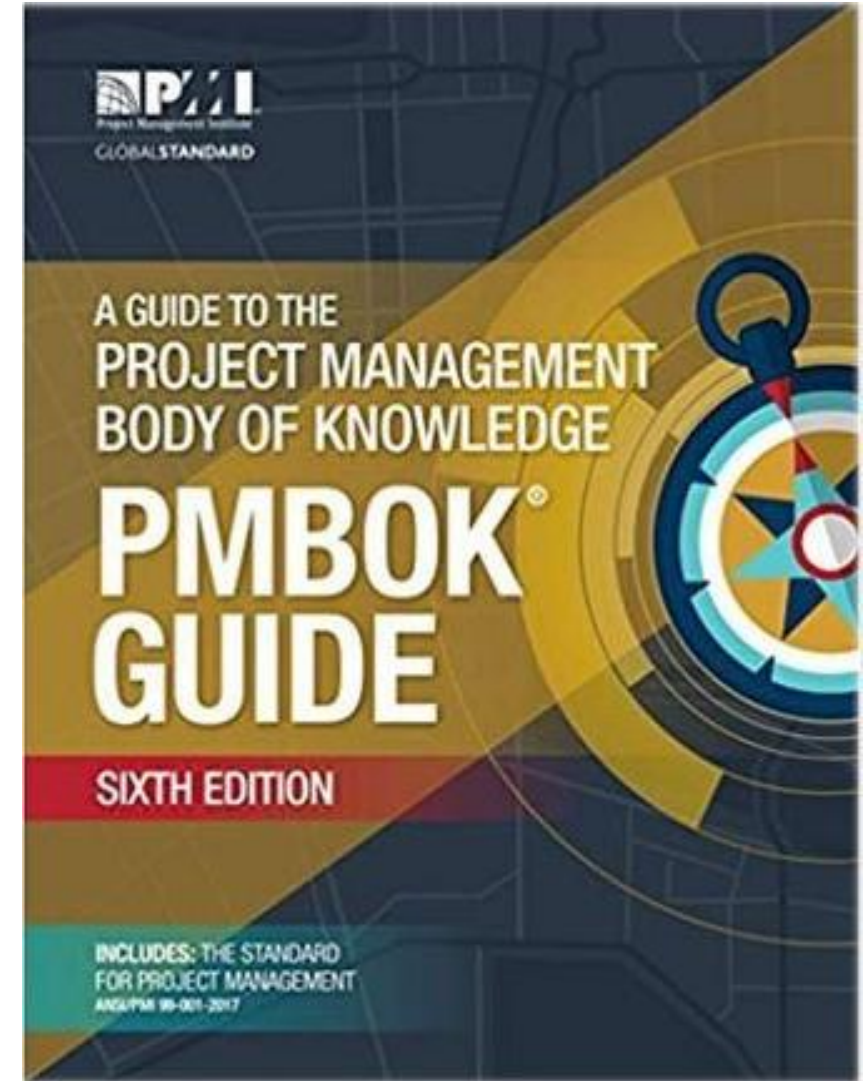
Who?

Difference between Project and Product?

- Discuss with your neighbors for 5 minutes and list out the differences.

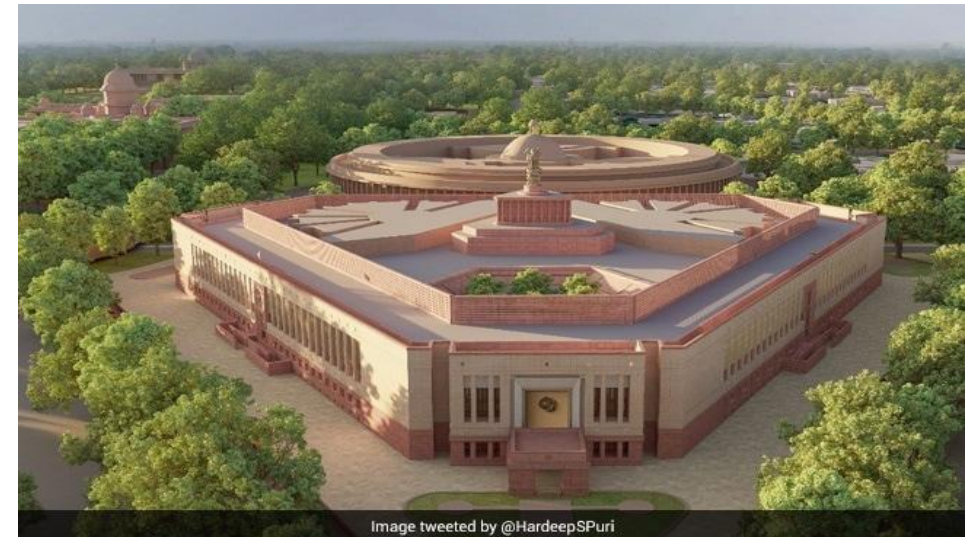
What is a Project?

- Project is a **temporary endeavour**
- Project has a definite **beginning and an end**
- Used to create **unique** product, service or results
- A building or facility is **produced**
- **Service** – design, planning, execution



Construction Project? – Unique features

- **One-time** activity
- Highly **complex**
- Costly and time consuming
- Multiple & unique **combination** of stakeholders
- **Unskilled** workforce



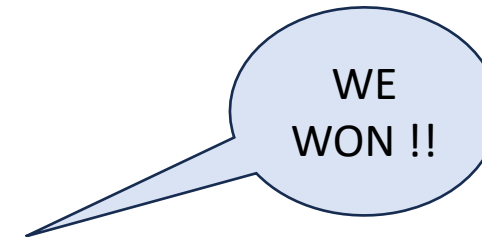
Some features which no one wants to talk about



Time overrun

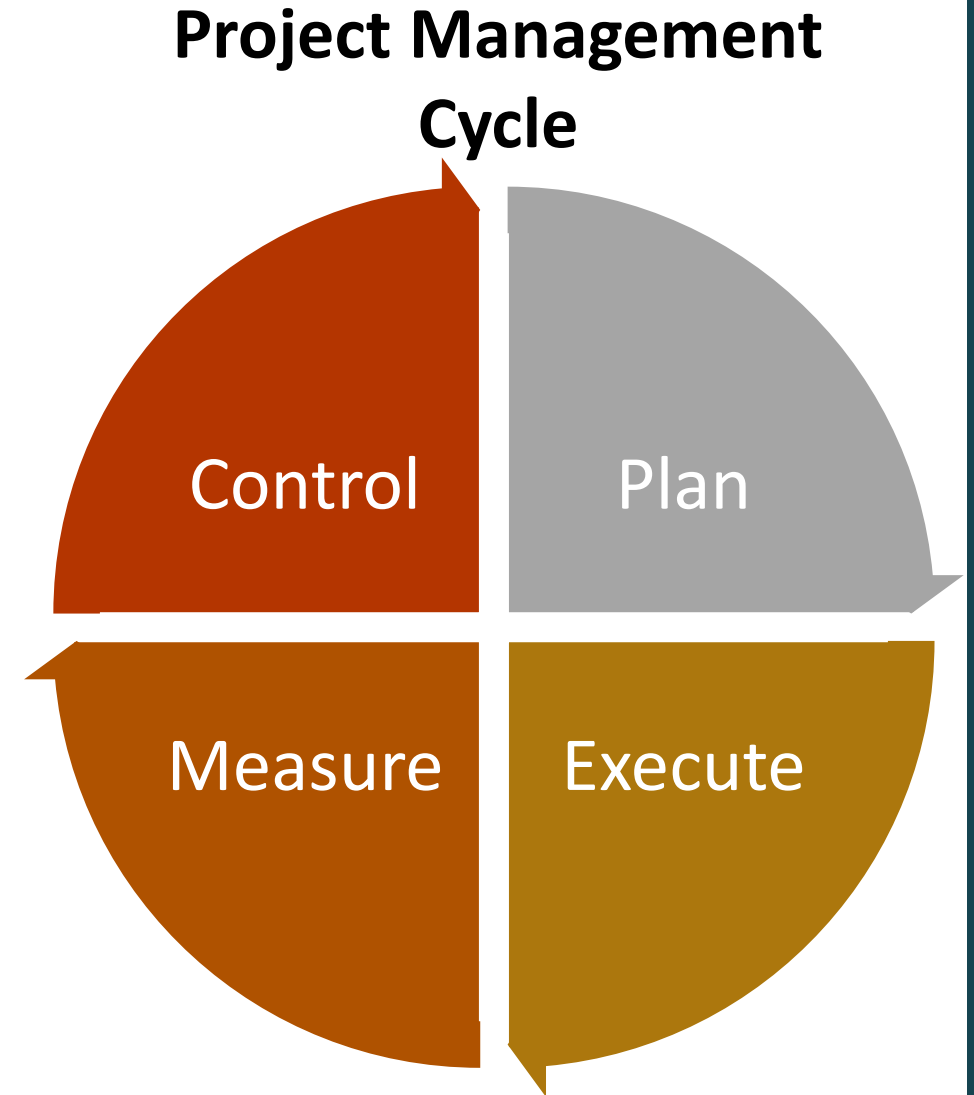
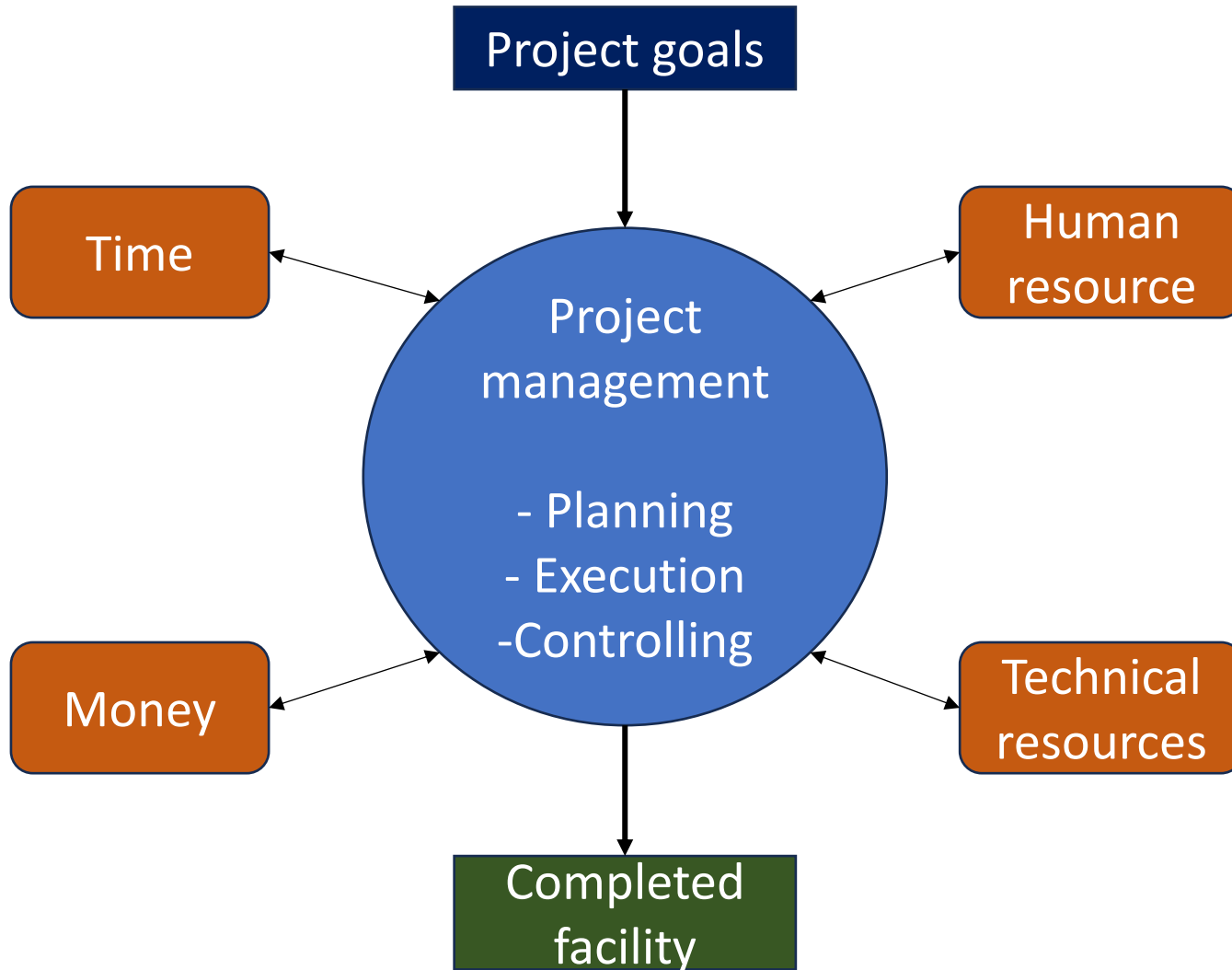


Cost overrun



Compromising the Scope/Content

Role of Project Management



Importance of Construction Project Management

- **Less than 20%** of construction project meet time, cost, safety, and quality goals
- Managing project $\neq$ managing steady state organization
- Construction utilizes lot of **natural scarce resources** and directly impacts the **environment**

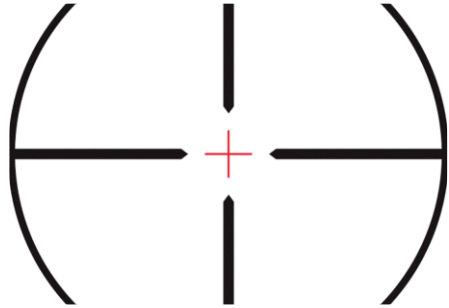


1632 - 22 years



2018 – 3.5 years

Project Management – Nine Knowledge Areas



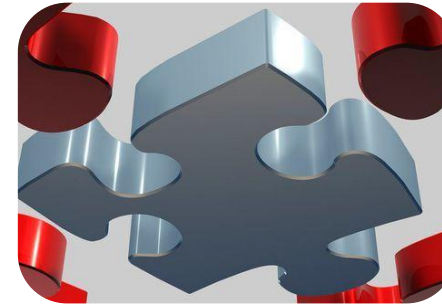
Scope
management



Time
management



Cost
management



Integration
management



Quality
management



Human
resource
management



Communication
management



Risk
management



Procurement
management

Project life cycle & Stakeholders

Construction Projects - Stakeholders



Owner (Client)



Architect



Engineer
(Consultant)



Construction contractor



Vendors/Supplier



Lawyer, Insurer etc.

Construction Project Life-cycle

Pre-project phase

- Idea or Initiation
- Project concepts
- Feasibility
- Conceptual drawings
- Project strategy
- Estimate & approval

Project phase

- Design phase
- Tendering phase
- Execution or construction phase
- Closure

Post-project phase

- Utilization phase
- Operation & Maintenance
- Close-down (dismantling & disposing)

Planning vs Scheduling?

- What is the difference between Planning vs Scheduling?

Planning

- PMBOK defines planning as “Those **processes performed** to **establish** the total **scope** of the effort, **define** and refine the **objectives**, and **develop** the **course of action** required to attain those objectives”.
- Project planning has been defined as “the **process of choosing the one method and order of work to be adopted for a project from all the various ways and sequences** in which it could be done” (Antill and Woodhead 1990, p. 8; Callahan, Quackenbush, and Rowings 1992)



Project Management vs Project Control ?

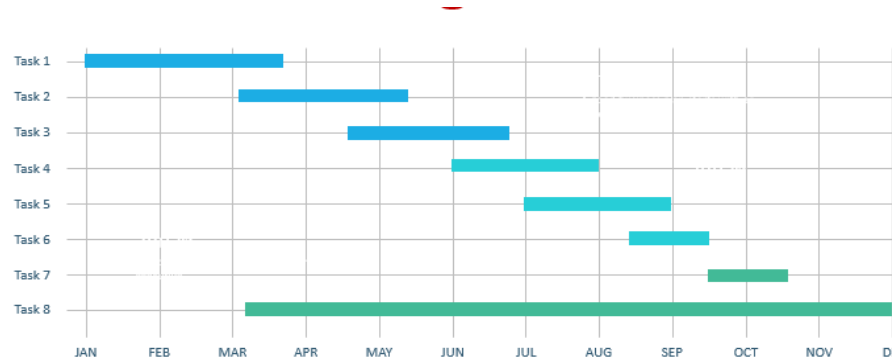
- **Project management** - defines **how project** will be **executed, monitored, and controlled**
- The objective of a **project management plan** is to **define the approach** to be used by the project team **to deliver the intended scope** of the project.”
(PMBOK 2008)
- **Project control** – Once project starts, **few aspects can deviate** from the plan. Managing them and **taking corrective action** (if needed) is project control.

Project vs Program or Programme vs Portfolio?

Types of Project Plans?



Material Plan



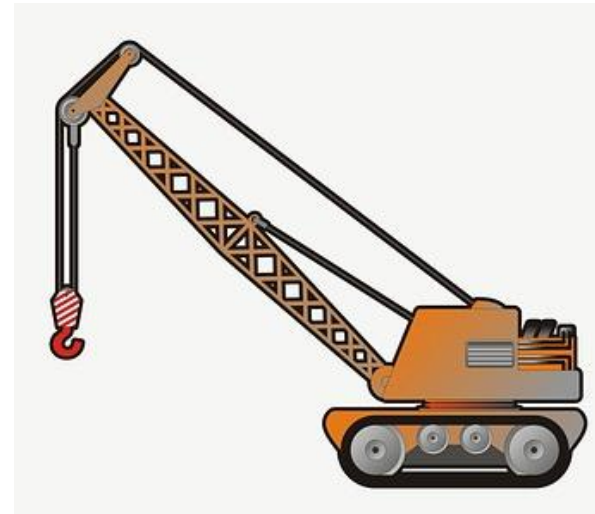
Time Plan (Schedule)



Workforce planning



Cost Plan (Finance)



P&M Plan

Discussion

What is more important for a project manager? - Managing Time or Managing cost?

- Discuss with your neighbors and answer
- Is managing time & cost inter-related? If so, can taking care of one will automatically take care of another?
- Can you think of a scenario where project managers should focus more on ensuring timely completion of work? (Without worrying about the total project cost)

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Scheduling of Projects – Introduction

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Why Schedule Projects?

Knowledge Required for Scheduling



Planning tools &
software

Knowledge of principles
of scheduling and
project management



Knowledge of specific
technical field

Activity, Duration, WBS

Work-breakdown structure

- Breaking down of project into “manageable chunks”
- WBS represents “family-tree” of activities. (Useful for large projects)
- WBS provides common framework for planning, scheduling, cost estimating, budgeting, monitoring, reporting, directing, and controlling
- Typically,
Project > Sub-project > Work-packages > Activities & Task

Work-breakdown structure

Definition as per PMBOK:

*“a **deliverable-oriented** hierarchical decomposition of the work to be executed by the project team to accomplish the project objectives and create the required deliverables.*

*It organizes and defines the total scope of the project. Each **descending level represents an increasingly detailed definition** of the project work. The WBS is decomposed into work packages. The deliverable orientation of the hierarchy includes both internal and external deliverables.”*

How to breakdown projects?

- No absolutely correct or incorrect way
- Always prefer a **balanced approach**
- The tasks/activities should be a **reasonable number**
- Activities should be **easy to measure and control without being overly detailed.**





THANK YOU